

# Manufacturing Operations & Plant Performance Overview

This report provides an executive-level overview of overall manufacturing performance, including production efficiency, downtime, quality performance, waste generation, and operational losses.

Period Covered: 1st Jan 2026 - 31st March 2026

OEE

58%

Availability

66%

Performance

91%

Quality

98%

Total Production Output (MT)

417585

Total Downtime in Hours

4,101

Total Production Time

7,820

Production Efficiency

91%

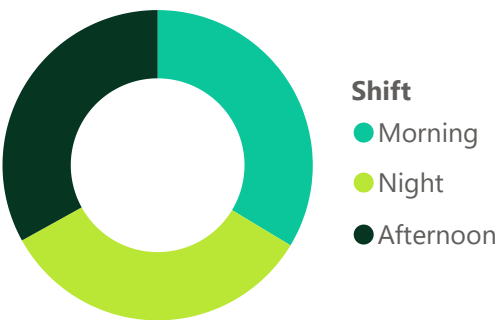
Production Variance

9%

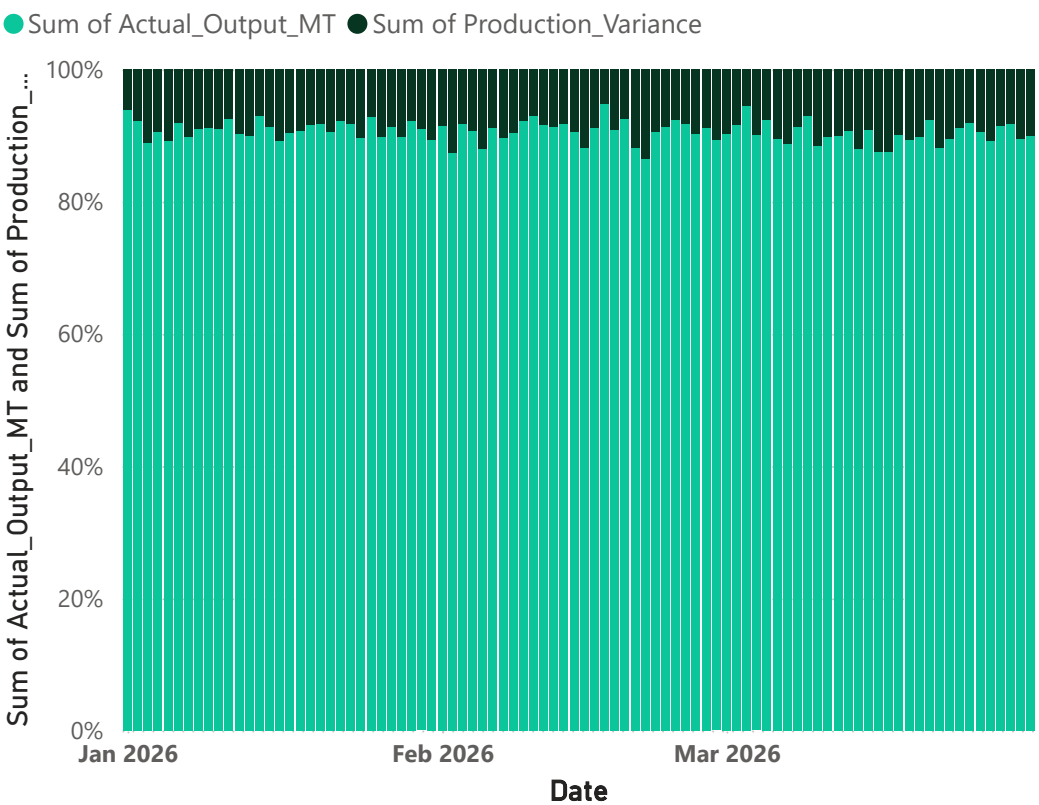
Quality Pass Rate

94%

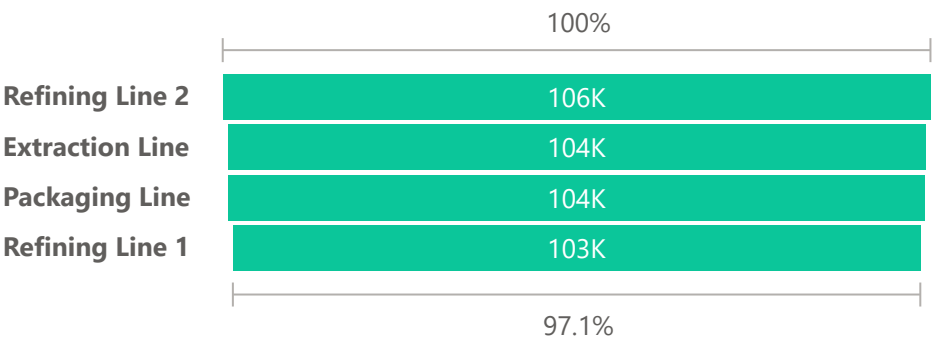
OEE by Shift



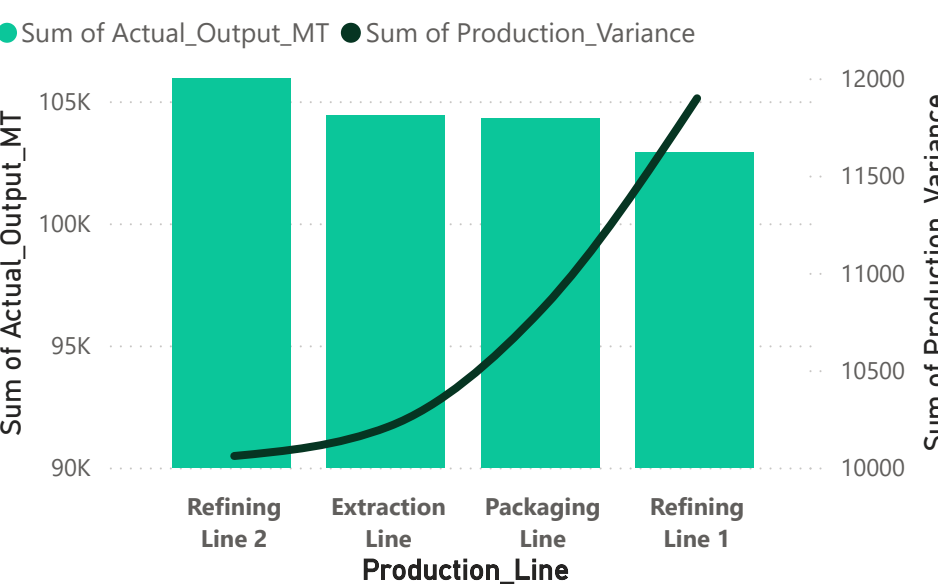
Production Output and Production Variance by Date



Production Output by Production\_Line



Production Output and Production Variance by Production\_Line



Date  
All

Capacity and Loss

Downtime and Maintenance

Quality Analytics

Shift Performance

Predictive Analysis

# Capacity Recovery & Production Loss Analysis

Date

All

Operator\_ID

☐ OP100

☐ OP101

☐ OP102

☐ OP103

☐ OP104

☐ OP105

☐ OP106

☐ OP107

☐ OP108

☐ OP109

☐ OP110

☐ OP111

☐ OP112

☐ OP113

☐ OP114

☐ OP115

Shift

☐ Afternoon

☐ Morning

☐ Night

This page analyzes the gap between planned and actual production output to uncover hidden production losses and recoverable manufacturing capacity.

The analysis helps identify underperforming production lines, unstable shifts, and operational bottlenecks affecting output efficiency.

Period Covered 1st Jan 2026 - 31st March 2026

91%

Capacity Utilization %

-43K

Variance MT

43K

Recoverable Capacity

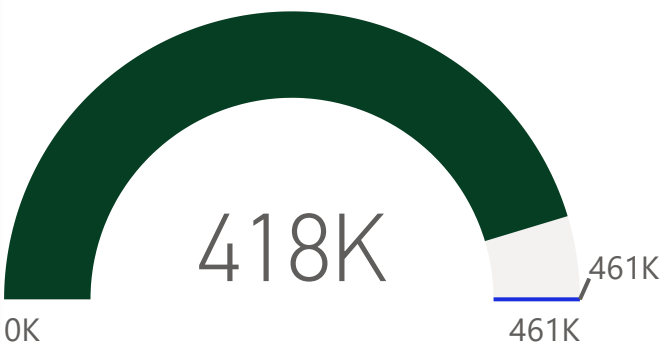
9.44

Average of Production\_Loss(%)

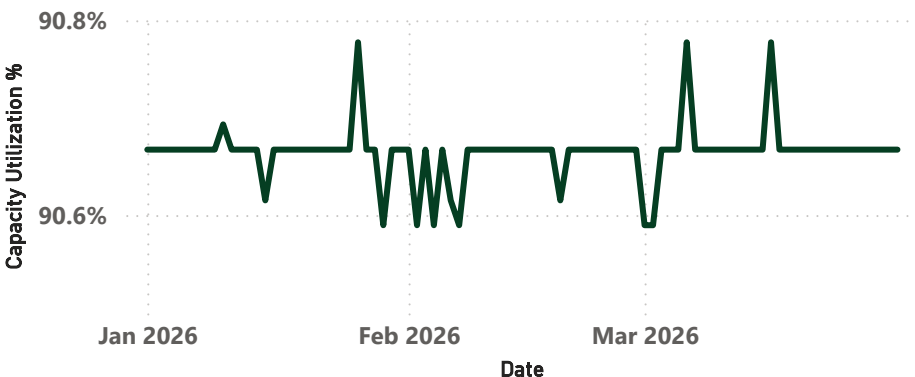
9.33%

Capacity Loss %

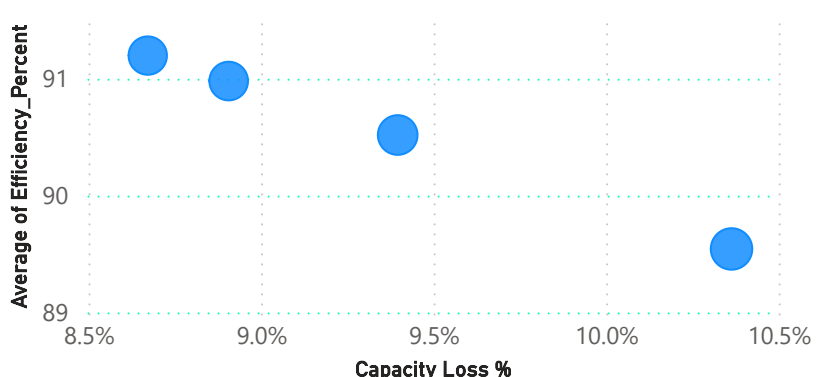
Production Achievement %



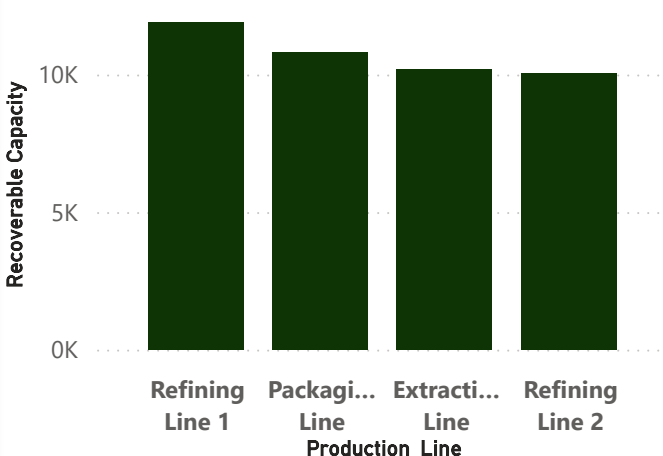
Capacity Utilization % by Date



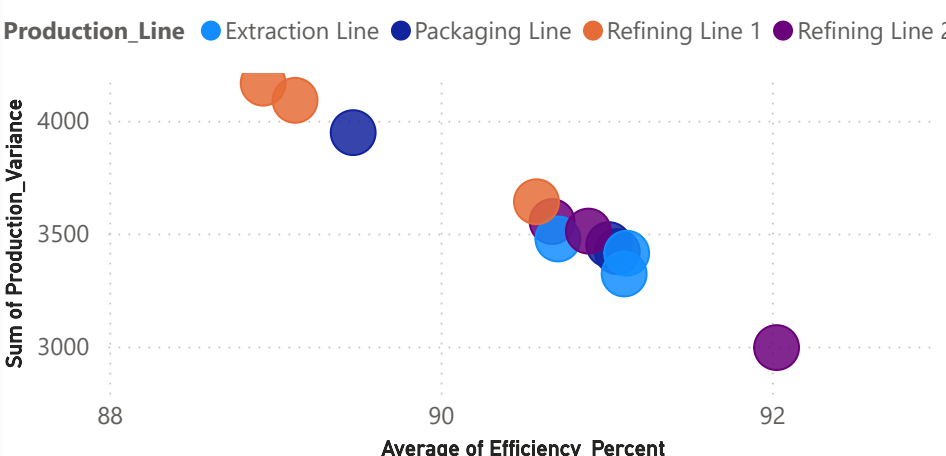
Capacity Loss vs Efficiency by Production Line



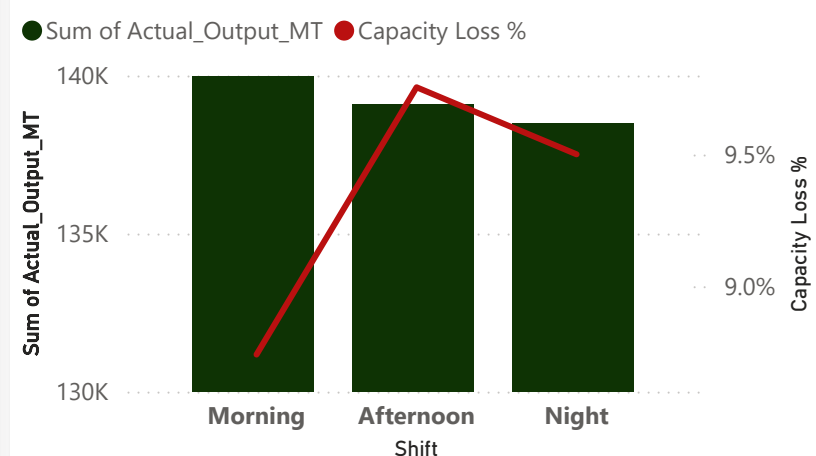
Recoverable Capacity by Production\_Line



Operational Performance by Shift & Line



Production Output and Capacity Loss % by Shift



# Downtime & Maintenance Intelligence Center

This page investigates machine downtime behavior, maintenance response effectiveness, and recurring operational disruptions across the plant. It helps identify the primary causes of production stoppages, high-risk machines, and maintenance inefficiencies contributing to productivity losses.

Period Covered: 1st Jan 2026 - 31st March 2026

59.04

Avg Downtime Per Day

5314

Downtime\_Incidents

3.40

Avg Daily Downtime

4,101

Downtime (Hrs)

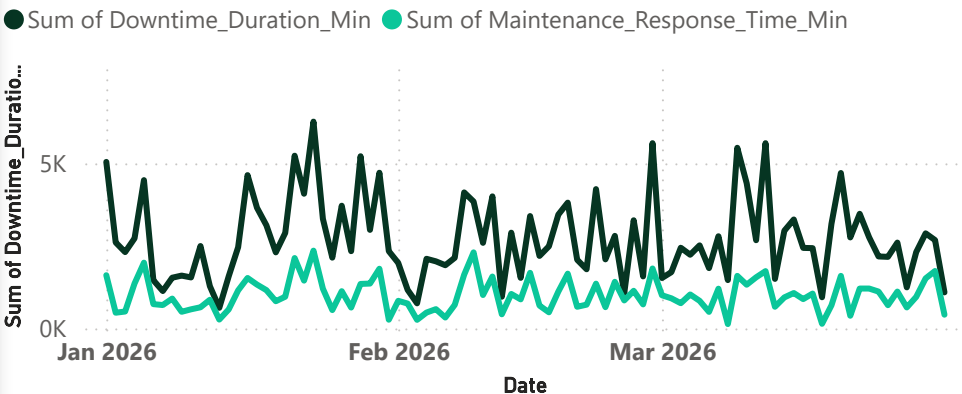
91332

Maintenance Response Time (...)

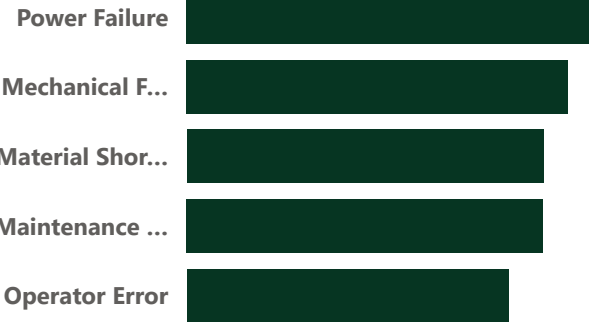
Production Time vs Downtime



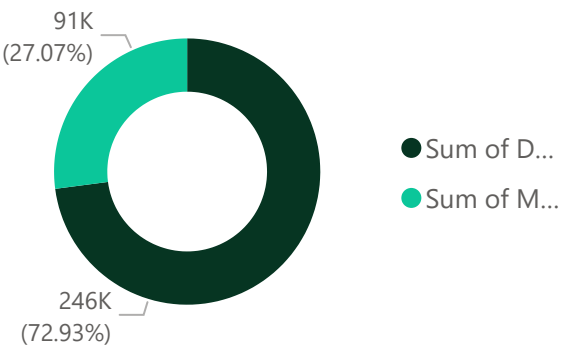
Downtime and Maintenance Response Time by Date



Downtime\_Causes

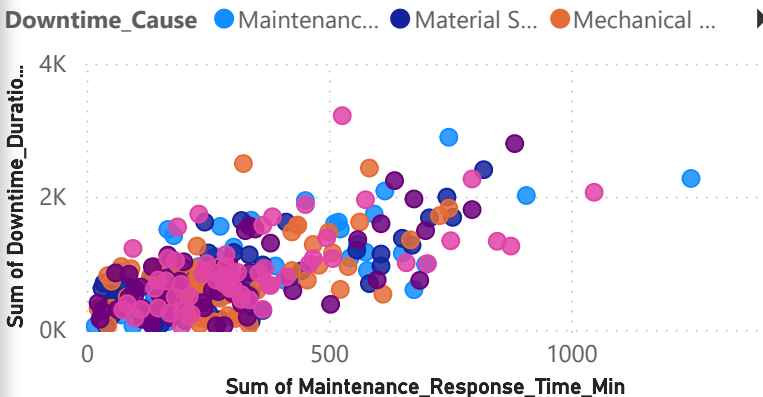


Downtime vs Maintenance Response Time



| Machine_ID | Total Downtime | Maintenance Response Time | Machine Risk Sco |
|------------|----------------|---------------------------|------------------|
| Filler     | 46767          | 17989                     | 31,994.          |
| Conveyor   | 43327          | 14460                     | 30,274.          |
| Deodorizer | 42096          | 15443                     | 29,658.          |
| Extractor  | 41557          | 15883                     | 29,390.          |
| Boiler     | 38079          | 13134                     | 27,649.          |
| Refinery   | 34209          | 14423                     | 25,716.          |

Maintenance Response Time and Downtime\_Duration by Date and Downtime Causes



# Quality, Waste & Rework Analytics

This page focuses on manufacturing quality performance by analyzing defects, waste generation, and rework activity across production operations. The report helps uncover hidden quality costs, process inconsistencies, and operational inefficiencies affecting usable output and production profitability.

Period Covered: 1st Jan 2026 - 31st March 2025

93.83

Average of Quality\_Pass\_Rate

663.89

Total Waste\_MT

51%

Rework Recovery Rate

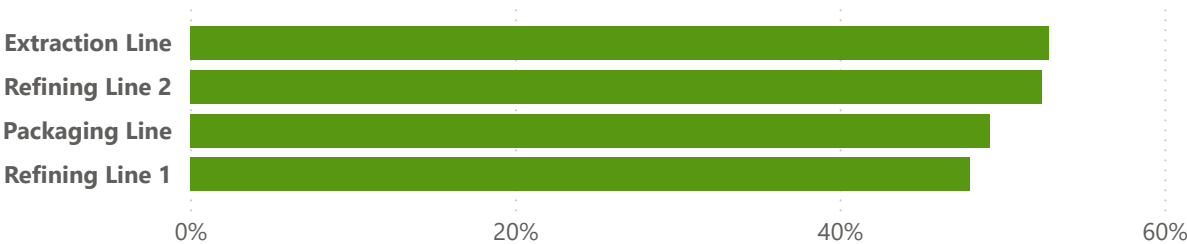
7569

Total Defective\_Units

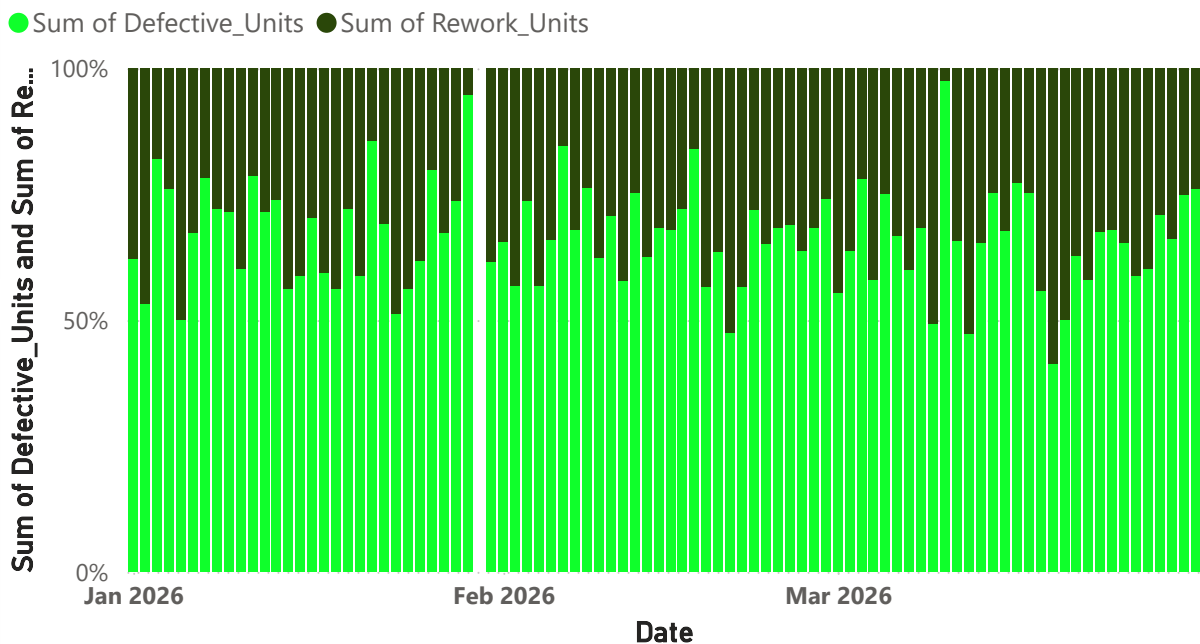
3840

Total Rework\_Units

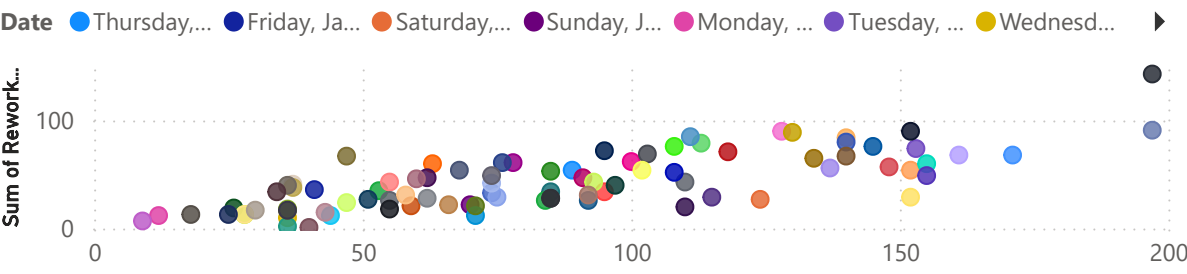
Rework Recovery Rate by Production\_Line



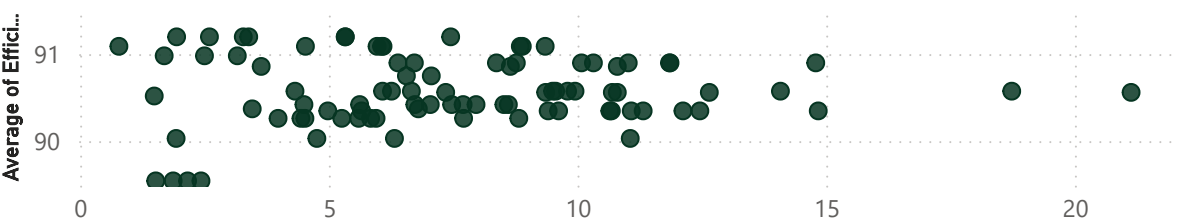
Defective\_Units and Rework\_Units by Date



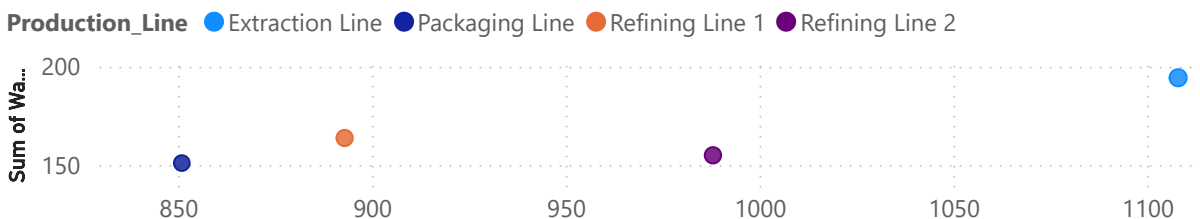
Defective\_Units, Rework\_Units and Waste\_MT by Date



Waste\_MT and Efficiency by Date



Rework\_Units, Waste\_MT, and Defective\_units by Production Line



Date

All

Production\_Line

Extraction Line

Packaging Line

Refining Line 1

Refining Line 2

# Shift Performance & Operational Risk Analysis

Date

All

Operator\_ID

☐ OP100

☐ OP101

☐ OP102

☐ OP103

☐ OP104

☐ OP105

☐ OP106

☐ OP107

☐ OP108

☐ OP109

☐ OP110

☐ OP111

☐ OP112

☐ OP113

☐ OP114

☐ OP115

☐ OP116

☐ OP117

☐ OP118

☐ OP119

This page evaluates operational stability across shifts, production lines, and machines using risk-based manufacturing analytics.

The analysis helps management identify high-risk operational areas, unstable production periods, and priority improvement opportunities requiring immediate attention.

Period Covered: 1st Jan 2026 - 31st March 2026

Morning

Best Shift

Afternoon

Least Performing Shift

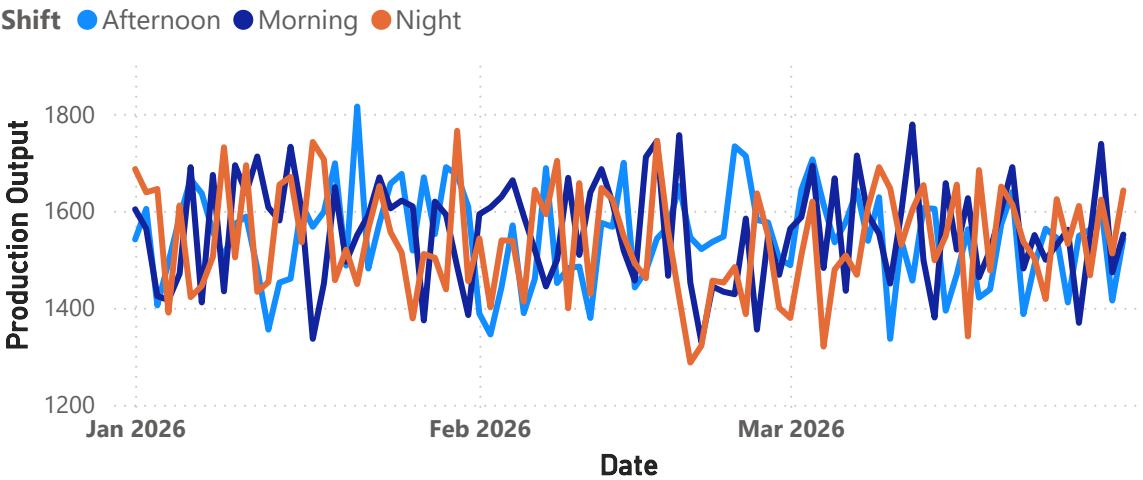
Extraction Line

Highest Waste Line

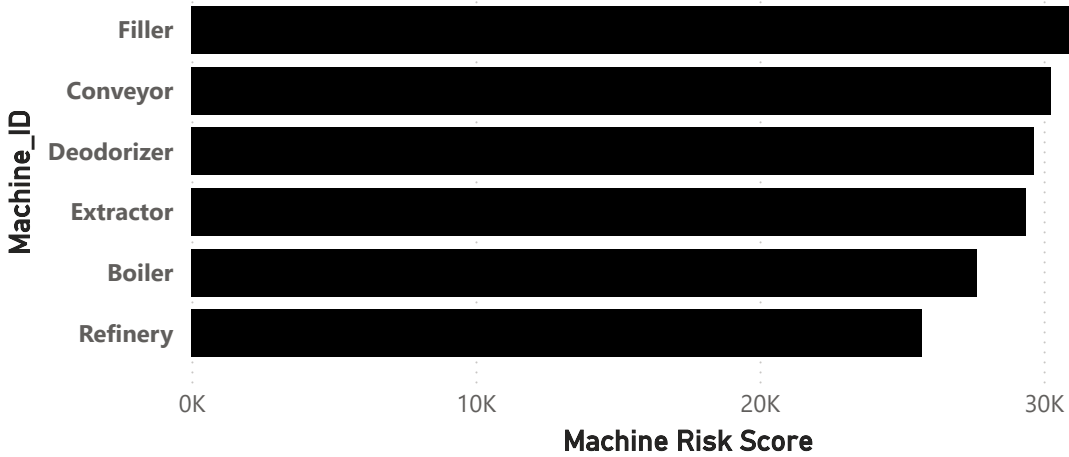
Packaging Line

Lowest Waste Line

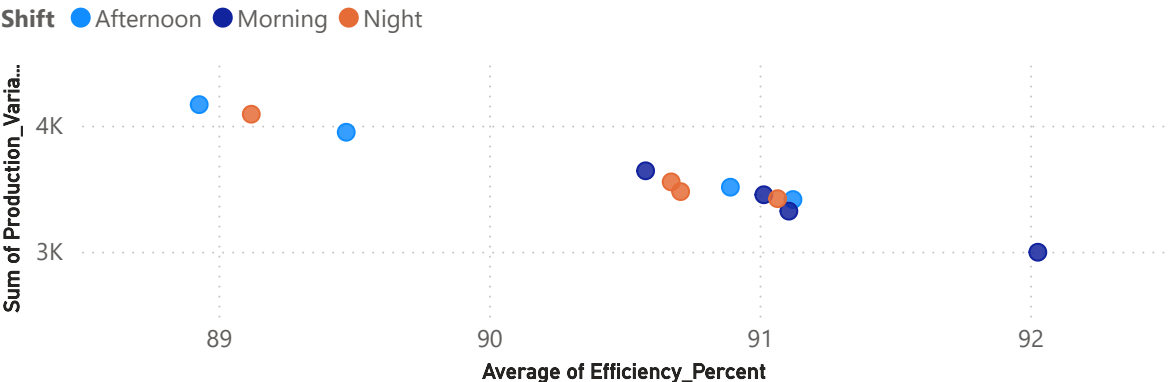
Production Output by Date and Shift



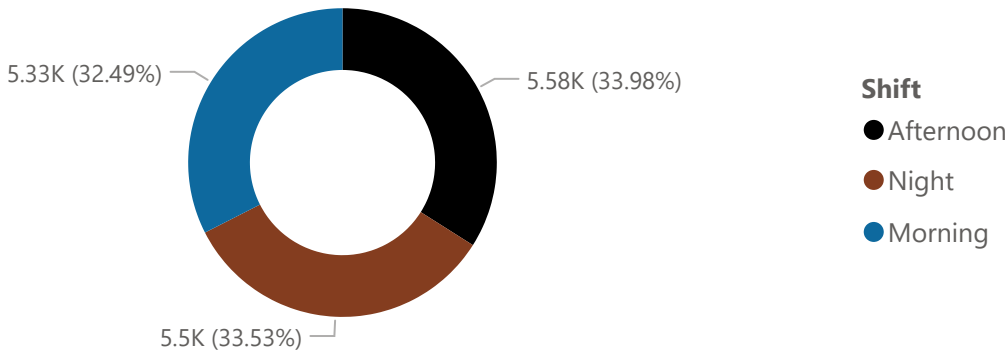
Machine Risk Score by Machine\_ID



Efficiency and Production loss by Production\_Line and Shift



Risk Score by Shift



Date

All

Production\_Line

☐ Extraction Line

☐ Packaging Line

☐ Refining Line 1

☐ Refining Line 2

Shift

☐ Afternoon

☐ Morning

☐ Night

Downtime\_Cause

☐ Maintenance Delay

☐ Material Shortage

☐ Mechanical Failure

☐ Operator Error

☐ Power Failure

Machine\_ID

☐ Boiler

☐ Conveyor

☐ Deodorizer

☐ Extractor

☐ Filler

☐ Refinery

# Forecasting Production, Quality & Operational Performance

This analysis provides a forward-looking view of manufacturing performance, highlighting potential shifts in production, downtime, quality, and operational efficiency to support proactive operational planning.

Period Covered: 1st Jan - 31st March 2026

